

TTEC Digital Development Project

The expectation is for you to build this out as you would a production project for a client, only on a small scale (tests, error handling, etc.). Where you don't have the time to implement something, add comments to your code or documentation on how you would have changed or added to your implementation in the "real world". You are allowed and encouraged to use AI but be prepared to defend your coding decisions. Not understanding the AI's choices and submitting sloppy code will not be rewarded.

Background

Acme Financial Services is launching a new Amazon Connect contact center to better serve their customers. Callers need the ability to dial in, provide their account number, and hear their current balance read back to them. Additionally, the company wants to generate vanity phone number options for their support line to make it easier for customers to remember and call back. Your job is to build the backend services and contact flow that power this experience.

Exercise

1. Create a Lambda function that converts a provided phone number to vanity numbers and save the best 5 resulting vanity numbers and the caller's number in a DynamoDB table. "Best" is defined as you see fit - explain your thoughts.
2. Implement a Lambda function as part of the AWS Connect Customer contact flow which calls the provided TTEC API and looks up a caller provided account number. Return the balance and firstName from the account in the API. Address the caller by name and read the balance.
3. Create an Amazon Connect Customer contact flow that accepts account number from the user. Look that account number up against the provided API. Takes the phone number assigned to that account from the API and use that as the phone number in the vanity number lambda. Finally, say the 3 vanity possibilities that come back from the Lambda function.
4. Build a IaC package with AWS CDK or Terraform to allow a user, or assignment reviewer, to deploy your Lambda, Lambda association, contact flow, DynamoDB table, and anything else you'd like to add into their own AWS Account/Amazon Connect instance.
5. **BONUS:** Provide an MCP for the team to review the results from DynamoDB
6. Writing and Documentation
 - a. Record your reasons for implementing the solution the way you did, struggles you faced and problems you overcame.
 - b. What shortcuts did you take that would be a bad practice in production?
 - c. What would you have done with more time? We know you have a life.
 - d. What other considerations would you make before making our toy app into something that would be ready for high volumes of traffic, potential attacks from bad folks, etc.
 - e. Please include an architecture diagram.
 - f. Please provide a screenshot of the flow from Connect

Deliverables

- A Git repo with all code and documentation. Minimum below
 - 2 Lambdas
 - 1 CDK / Terraform
 - 1 Call flow
 - Documentation
- A working Amazon Connect phone number to test in your environment

TTECDigital API information

Base URL

<https://d3ot99cmewuv1b.cloudfront.net>

No authentication required.

Endpoints

GET /allAccountNumbers

Returns all valid account numbers as a JSON array of strings. Use these to test your contact flow.

GET /accountLookup?accountNumber={id}

Returns account details for the given account number.

Response:

```
{
  "accountNumber": "847",
  "balance": 4521.3,
  "firstName": "Jane",
  "lastName": "Smith",
  "phoneNumber": "+14783472937",
  "status": "active"
}
```

Returns 404 if the account number does not exist.

Resources

Resource	URL
OpenAPI Spec	https://d3ot99cmewuv1b.cloudfront.net/docs/openapi.yaml
Postman Collection	https://d3ot99cmewuv1b.cloudfront.net/docs/postman/coding-assignment-api.postman_collection.json